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The Editor
Computer Methods and Programs in Biomedicine

Dear Editor,

We submit the manuscript "Certified adaptive examination of peripheral blood smears: when information-gain stopping fails, and what can be guaranteed instead" for consideration as an original research article.

Why this journal. The manuscript develops and evaluates computational methods, adaptive sampling designs and sequential statistical tests, for the automated examination of peripheral blood smears, and releases the complete software with a pre-registration audit trail and a reproducibility package, within the journal's scope of computing methodology and software for biomedicine.

What is known. Automated smear analysis increasingly decides which fields to examine and when to stop; manual differentials are imprecise for rare cell types, and information-saturation stopping has recently been proposed for slide-level image analysis. Design-based, anytime-valid tests for partly examined finite populations exist in election auditing but had not been evaluated for smear examination, and whether their assumptions hold on real smear data was unknown.

What this study adds.

- A pre-registered evaluation showing that information-saturation stopping does not certify unexamined material: its stopping point barely changed with rarity, and with one critical field among 200 ten-cell fields it missed 76% of positive specimens, against 85% for uniform sampling of the same size.
- A minimax lower bound showing that certifying the absence of rare cells requires examining nearly the whole specimen for every strategy without location information, including any halting rule that ignores verified labels.
- A demonstration that a blast percentage above a diagnostic threshold can be certified with finite-sample, anytime-valid guarantees, with a cost in expert-verified cells governed mainly by the margin to the threshold; near the threshold a conventional 200-cell count was wrong in up to 27% of specimens.
- An empirical test of the assumptions on two external datasets and on exhaustively annotated bone-marrow regions, showing that validity depends on the sampling design, the unit of analysis and its weighting, and the reliability of verification, whereas classifier quality affects efficiency in both directions: against uniform sampling with the ALPHA test, classifier-driven sampling needed 0.45–0.75 times the verified cells in distribution and, at margins of 0.05–0.10, 1.3–3.0 times under acquisition shift, an excess that a uniform mixture component limits.

The manuscript has not been published elsewhere, in whole or in part, and is not under consideration by another journal. Both authors have approved the submission. The study used only publicly available, de-identified datasets. [Preprint server and DOI, if posted.]

Yours sincerely,

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